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Efficiency and Redistribution in Environmental Policy: An Equilibrium Analysis of Agricultural Supply Chains

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1 Big picture

- **Question.** How should environmental policy be designed when the externality is created upstream by millions of atomistic farmers, but feasible regulation is easier to impose downstream on a concentrated sector of agribusiness intermediaries?
- **Setting.** The paper studies the **South American agricultural sector**, focusing mainly on **Argentina and Brazil** and combining county-level land use, output, prices, trade flows, and carbon-stock data from **1995–2017**.
- **Why the question matters.** Agriculture accounts for a large share of global emissions, and in South America the key margins—**deforestation**, **commodity choice**, and **production location**—all sit upstream at the farm level rather than at the concentrated downstream stage.
- **Core challenge.** A first-best carbon tax on farmers is theoretically clean, but in practice it is difficult to enforce and politically costly. The realistic alternatives are **second-best policies** implemented further downstream, where the Pigouvian signal must pass through an imperfect supply chain before it reaches farmers.
- **Main conceptual contribution.** The paper builds a **county-level equilibrium model** of agricultural supply chains in which a nested land-use model on the supply side is combined with a layer of **oligopsonistic agribusiness buyers** on the demand side.
- **Main empirical finding on supply.** Farmers in frontier regions near dense forests have **lower supply elasticities**: they are less responsive both in switching across commodities and in moving land between agriculture and forest.
- **Main market-power finding.** Because oligopsony markdowns are wider when supply is inelastic, **agribusiness market power is strongest in remote, carbon-intensive frontier regions**. Thus the environmental externality and market power are positively correlated across space.
- **Main downstream-policy result.** A feasible **uniform downstream tax** reduces emissions on average, but it is **poorly targeted**: production falls least in the high-emissions frontier regions where social damage is largest.
- **Main mechanism.** Downstream policy works only through **pass-through** to farm-gate prices. Monopsony power weakens this pass-through exactly where supply is least elastic, so the Pigouvian signal is most eroded where environmental cost is highest.
- **Distributional result.** The downstream tax is **regressive on the supply side**: farm-gate prices fall most in poorer frontier regions, so poor farmers bear a larger implicit tax burden than rich farmers.
- **Targeting-vs.-scale result.** A **targeted upstream intervention** (modeled as a forest subsidy/conservation-zone policy) is much better targeted but limited in scale by enforcement costs. A **coarsely targeted downstream policy** based on macroregion certification preserves scale and already improves targeting substantially.
- **Competition result.** Market power is not unambiguously good or bad for the environment. What matters is the **sign of the correlation** between market power and emissions intensity.

In this setting it is positive, so market power worsens policy transmission.

- **Welfare result.** Relative to the first best, the paper’s feasible second-best policies recover only about **29–38%** of first-best emissions reductions and **26–36%** of first-best welfare gains.
- **Economic takeaway.** When regulation at the source is infeasible, downstream environmental policy can be both **inefficiently targeted** and **regressive**. The structure of the supply chain—especially market power between the regulator and the emitter—is therefore central for environmental policy design.

2 Data and measurement (key objects)

2.1 Geographic unit, time frame, and commodities

The paper constructs a **county-level panel** for Argentina and Brazil. Argentine observations are **departments/partidos**; Brazilian observations are **AMCs** (*áreas mínimas comparáveis*) so that units are time-consistent despite municipal boundary changes.

The core sample spans **1995–2017**. Because many key data come from decadal agricultural censuses, the paper interprets much of the time variation as **long-run adjustment** rather than short-run annual responses.

The main commodities are:

- **beef cattle / pasture**,
- **soybeans**,
- **maize**,
- **wheat**,
- **rice**,
- **sunflower**,
- **sugarcane**.

Beef, soybeans, and maize account for roughly **85% of agricultural land**, and **pasture alone accounts for about 70%**.

Interpretation. The model is broad enough to allow land reallocation across multiple commodities, but the quantitative policy action is dominated by the beef-pasture margin.

2.2 Supply-side data

The supply side combines:

- county-level **land use** from agricultural censuses in Argentina and Brazil;
- county-level **agricultural output** from censuses and official production surveys;
- high-resolution **agronomic productivity** from **FAO-GAEZ**, which gives potential yields predicted by agroclimatic fundamentals;

- realized yields used together with FAO-GAEZ to calibrate location-commodity productivity.

Interpretation. The paper needs both land allocation and physical productivity because the key supply object is not just how much farmers produce, but also **where** land is cleared and **which commodity** it is allocated to.

2.3 Demand-side and trade-flow data

To connect county production to downstream buyers and final markets, the paper uses:

- **FAOSTAT** for national trade flows,
- **Trase** domestic supply-chain data to map annual flows from source counties to destination markets and to agribusiness intermediaries,
- for Argentine beef, meat-packer procurement records from **DA-MAGYP**.

Trase is particularly important because it provides both:

- the identity of **destination markets**, and
- the identity of the **agribusiness firms** intermediating the transactions.

Interpretation. This is what lets the paper move from a standard trade model to a **supply-chain model with buyer concentration**.

2.4 Prices, concentration, and markdowns

The paper uses two key price concepts:

- **farm-gate prices**, constructed from local production values and quantities;
- **destination / FOB prices**, constructed from Trase trade values and quantities.

The paper first builds an **accounting markdown**:

$$\text{Accounting markdown}_{it} = \frac{\text{farm-gate price}_{it}}{\text{agribusiness FOB price}_{it}}. \quad (1)$$

It also measures buyer concentration with concentration ratios such as **CR-1** and **CR-3** in each local upstream market.

A related control is a standard **market-access** measure,

$$MA_i = \sum_j s_j d_{ij}^{-1}, \quad (2)$$

where d_{ij} is distance to hub j and s_j weights hub size.

Interpretation. The accounting markdown is not the paper's structural market-power measure, but it provides a transparent first pass: are farm-gate prices relatively lower exactly where agribusiness buyers are more concentrated?

2.5 Emissions objects

The paper decomposes agricultural emissions into two components:

- **land-use-change (LUC) emissions**, driven by deforestation and based on carbon-stock maps from Spawn and Gibbs (2020);
- **non-LUC on-farm emissions**, based on commodity-specific life-cycle assessments from Poore and Nemecek (2018).

The location-level LUC intensity is measured as

$$e_i^{\text{LUC}} = \text{CO}_2\text{e stock per hectare of land in location } i, \quad (3)$$

while the commodity-level on-farm intensity is

$$e_i^{c,\text{NLUC}} = \text{tonnes of non-LUC CO}_2\text{e per tonne of commodity } c. \quad (4)$$

The baseline **social cost of carbon (SCC)** used in the counterfactuals is **US\$50 per tonne of CO₂e**.

Interpretation. The paper is not just about how much agricultural output changes. It is about whether policy suppresses output in the **right commodities** and the **right places**.

2.6 Frontier versus core regions

The paper allows key supply responses to differ between **frontier** and **core** agricultural regions.

- In Brazil, the frontier is **Amazonia Legal**.
- In Argentina, the core is the **zona núcleo / Pampeana** region.

Interpretation. This heterogeneity matters because frontier regions are exactly where carbon stocks are high, market access is poor, and agribusiness buyer power is most likely to matter for policy transmission.

2.7 Key descriptive facts

The descriptive evidence establishes three core facts:

- **Fact 1:** More than **80%** of agricultural emissions arise upstream before the commodity leaves the farm gate, largely through land use change and on-farm sources; in South America the land-use share exceeds **70%**.
- **Fact 2:** Supply chains are **funnel shaped**. In Brazil there are roughly **2.46 million** ranching establishments, but only about **134** beef-buying agribusiness firms nationally; in the median local beef market, the **top three buyers account for 95%** of sourcing.
- **Fact 3:** Commodities **compete for land**. The South American soybean boom reallocated land away from pasture in core regions and indirectly pushed cattle toward frontier zones where forests are located.

Interpretation. These facts motivate the model structure exactly: upstream environmental choices, downstream concentration, and multi-commodity land competition all matter simultaneously.

3 Models and empirical specifications (all equations + interpretation)

3.1 Farmer production technology

At the field level, a farmer who rents land to produce commodity c in county i and period t uses a Cobb-Douglas technology:

$$Q_{it}^c(\omega) = \bar{q} [A_{it}^c(\omega) L_{it}^c(\omega)]^{\gamma_L^c} [H_{it}^c(\omega)]^{\gamma_H^c} [M_{it}^c(\omega)]^{\gamma_M^c}, \quad (5)$$

with

$$A_{it}^c(\omega) = A_{it}^c \exp(\varepsilon_{it}^c(\omega)), \quad (6)$$

and factor shares satisfying $\gamma_L^c + \gamma_H^c + \gamma_M^c = 1$.

Interpretation. Productivity has both a county-commodity component and a field-level idiosyncratic component. This field heterogeneity is what later generates smooth land reallocation rather than corner solutions.

The associated unit cost of production is

$$\kappa_{it}^c(\omega) = (w_{Hit})^{\gamma_H^c} (w_{Mit})^{\gamma_M^c} \left(\frac{r_{it}^c(\omega)}{A_{it}^c(\omega)} \right)^{\gamma_L^c}, \quad (7)$$

where $r_{it}^c(\omega)$ is land rent, w_{Hit} is the local wage, and w_{Mit} is the intermediate-input price.

Interpretation. Farm-gate prices affect output only indirectly through land rents, nonland factors, and the extensive land-allocation margin.

3.2 Landowner's land-use choice

Each field is owned by a landowner who chooses between natural land use N and an agricultural nest C . The field-level choice problem is

$$\max_{\{1, \dots, c, \dots, C, N\}} \left[r_{it}^1 e^{\varepsilon_{it}^1(\omega)}, \dots, r_{it}^c e^{\varepsilon_{it}^c(\omega)}, A_{it}^N e^{\varepsilon_{it}^N(\omega)} \right]. \quad (8)$$

Here A_{it}^N is the payoff from leaving the field in its natural state.

Interpretation. This is the paper's central environmental margin. Policies that directly protect forests operate through A_{it}^N ; policies that work through prices operate through the rents r_{it}^c .

3.3 Nested land shares and supply elasticities

With generalized extreme value shocks, the model implies a nested structure. Conditional on choosing agriculture, the share allocated to commodity c is

$$\pi_{it}^{c|C} = \frac{(r_{it}^c)^{\theta/\lambda}}{\sum_{c' \in C} (r_{it}^{c'})^{\theta/\lambda}}, \quad (9)$$

while the probability of choosing the agricultural nest is

$$\pi_{it}^C = \frac{(P_{it}^C)^\lambda}{(A_{it}^N)^\theta + (P_{it}^C)^\lambda}, \quad P_{it}^C \equiv \sum_{c' \in C} (r_{it}^{c'})^{\theta/\lambda}. \quad (10)$$

Interpretation.

- θ/λ governs **within-agriculture substitution** across commodities.
- λ governs **across-nest substitution** between agriculture and forest, that is, the **deforestation elasticity**.

Higher θ/λ means easier switching across crops; lower λ means lower responsiveness of land clearing to agricultural incentives.

The model then aggregates field decisions into county-level acreage and output. The paper derives a closed-form county supply equation (its eq. (3)) that depends on total land, land shares, prices, and factor shares.

Interpretation. For the counterfactuals, what matters is that supply is not one-dimensional: farmers respond by changing **how much land is agricultural** and **which agricultural use that land takes**.

3.4 Downstream agribusiness demand and the markdown equation

There are N_{it}^c identical agribusiness firms purchasing raw commodity c from source county i . Each firm can transform the raw commodity into a downstream product k and sell it at FOB price $\bar{p}_{it}^k$.

The key first-order condition is

$$p_{it}^c = \mu_{it}^c \bar{p}_{it}^k f'(q_{it}^c; k), \quad \mu_{it}^c = \left(1 + \frac{\partial \ln p_{it}^c}{\partial \ln Q_{it}^c} \frac{1}{N_{it}^c} \right)^{-1} < 1. \quad (11)$$

Interpretation.

- p_{it}^c is the farm-gate price received by farmers.
- $\bar{p}_{it}^k f'(\cdot)$ is the marginal revenue generated downstream.
- μ_{it}^c is the **markdown**: the share of marginal revenue passed upstream to farmers.

This is the paper's core market-power mechanism. When supply is more inelastic, the inverse-supply elasticity term is larger, so μ_{it}^c is smaller and the farm-gate price is marked down more.

3.5 Final demand

Final demand is modeled with a three-level CES system. Destination j 's demand for product k from county i in nation n is

$$C_{ijkt}^k = a_{ijkt}^k \left(\frac{p_{ijt}^k}{P_{njt}^k} \right)^{-\eta_l} a_{njt}^k \left(\frac{P_{njt}^k}{P_{jt}^k} \right)^{-\eta_m} a_{jt}^k \left(\frac{P_{jt}^k}{P_{jt}^k} \right)^{-\eta_u} \frac{X_{jt}}{P_{jt}^k}, \quad i \in n. \quad (12)$$

Interpretation. Consumers substitute:

- across **counties** within a nation with elasticity η_l ,
- across **source nations** with elasticity η_m ,
- across **products/commodities** with elasticity η_u .

This layered demand system is what lets the model evaluate both domestic reallocation and trade diversion under downstream environmental policy.

3.6 Equilibrium conditions

An equilibrium is a vector of farm-gate prices and local wages such that:

$$Q_{it}^c(p_{it}^c) = \sum_j C_{ijt}^c(p_{ijt}^k p_{ij}^{\tau_{ij}}) \quad \forall i, c, \quad (13)$$

and local agricultural labor supply equals local agricultural labor demand:

$$s_{H|it}(w_{Hit}, \bar{w}_{Hit})^{(\psi-1)/\psi} \bar{H}_{it} = \sum_c H_{it}^c(w_{Hit}, w_{Mit}, p_{it}^c). \quad (14)$$

Interpretation. The model is an equilibrium object, not a reduced-form passthrough regression. Any policy changes farm-gate prices, land use, labor allocation, trade flows, and emissions jointly.

3.7 Emissions accounting

Location-level emissions changes satisfy

$$\Delta E_i = \Delta E_i^{\text{LUC}} + \sum_c \Delta E_i^{c, \text{NLUC}}, \quad (15)$$

with

$$\Delta E_i^{\text{LUC}} = \Delta L_i^C e_i^{\text{LUC}}, \quad \Delta E_i^{c, \text{NLUC}} = \Delta Q_i^c e_i^{c, \text{NLUC}}. \quad (16)$$

Interpretation. Policies can reduce emissions by shrinking agricultural land, by shifting land into cleaner commodities, or by moving production to lower-carbon locations.

3.8 Supply-side estimating equations

To estimate the within-agriculture substitution elasticity, the paper uses the odds ratio between two commodities c and c' :

$$\ln\left(\frac{\pi_{it}^c}{\pi_{it}^{c'}}\right) = \frac{\theta}{\lambda} \ln\left(\frac{r_{it}^c}{r_{it}^{c'}}\right) = \frac{\theta}{\lambda} \ln\left(\frac{(p_{it}^c)^{1/\gamma_L^c} z_{it}^c}{(p_{it}^{c'})^{1/\gamma_L^{c'}} z_{it}^{c'}}\right) + u_{it}^{cc'}. \quad (17)$$

Interpretation. Relative land shares are related to relative farm-gate prices and relative agronomic productivity. The coefficient θ/λ is the structural elasticity of substitution across commodities within agricultural land.

To estimate substitution between agriculture and natural land use, the paper uses

$$\ln\left(\frac{\pi_{it}^C}{\pi_{it}^N}\right) = \lambda \ln(P_{it}^C) + u_{it}^N. \quad (18)$$

Interpretation. This equation identifies the **deforestation elasticity**. If λ is small, even large changes in agricultural returns do little to change the agriculture-vs.-forest split.

3.9 Demand-side estimating equations

The lower-level elasticity across counties is estimated from

$$\ln\left(\frac{X_{ijt}^c}{X_{njt}^c}\right) = (1 - \eta_l) \ln p_{ijt}^c + \iota_{njt}^c + \varepsilon_{ijt}^c. \quad (19)$$

The middle-level elasticity across source nations is estimated from

$$\ln\left(\frac{X_{njt}^c}{X_{jt}^c}\right) = (1 - \eta_m) \ln P_{njt}^c + \iota_{jt}^c + \varepsilon_{njt}^c. \quad (20)$$

The upper-level elasticity across commodities is estimated from

$$\ln\left(\frac{X_{jt}^c}{X_{jt}^c}\right) = (1 - \eta_u) \ln P_{jt}^c + \iota_{jt}^c + \varepsilon_{jt}^c. \quad (21)$$

Interpretation. Because the coefficient is $1 - \eta$, a more negative price coefficient means stronger substitution. These elasticities determine how final demand shifts across origins and commodities when environmental policy changes prices.

3.10 Policy comparison and first best

To compare a targeted upstream policy to an untargeted downstream one, the paper defines the **targeting premium**

$$T \equiv \Delta E_U - \Delta E_D, \quad (22)$$

and states that the upstream tool is preferred if

$$SCC \times \Delta E_U - C > SCC \times \Delta E_D \quad \Longleftrightarrow \quad C < SCC \times T, \quad (23)$$

where C is the extra enforcement cost of the upstream policy.

Interpretation. This converts the targeting-vs.-scale trade-off into one sufficient statistic: how much extra enforcement cost society should be willing to pay for targeting.

The first-best Pigouvian benchmark under perfect competition is

$$t_i^{c*} = SCC \times e_i^{c, \text{NLUC}}, \quad s_i^{N*} = SCC \times e_i^{\text{LUC}}. \quad (24)$$

Under imperfect competition, the output tax that decentralizes the first best becomes

$$t_i^c = t_i^{c*} - p_i^{c*} [1 - \mu_i^c(L_i^{c*})]. \quad (25)$$

Interpretation. This is Buchanan’s classic second-best logic. If market power is already depressing output, the optimal corrective tax is **smaller** than under perfect competition.

4 Identification and instruments (what makes the estimates credible)

4.1 What the paper is and is not identifying

The paper is not a reduced-form causal estimate of “the effect of concentration on markdowns.” In fact, the paper explicitly emphasizes that concentration, prices, and markdowns are all **equilibrium outcomes** determined by supply and demand primitives.

Instead, the identification problem is structural:

- estimate supply elasticities for land reallocation and deforestation;
- estimate final-demand substitution elasticities;
- combine these objects with the observed number of downstream buyers to infer markdowns;
- then use the full model for counterfactual policy evaluation.

Interpretation. Credibility comes from transparent sources of variation for elasticities, not from treating observed concentration as quasi-random.

4.2 Identification of within-nest supply substitution

Equation (7) is estimated with a **demand-shifter IV**. For each commodity c , the paper builds a Bartik-style instrument

$$IV_{it}^c = \sum_j s_{ij}^c d_{jt}^c, \quad (26)$$

where s_{ij}^c is county i ’s historical exposure to destination j and d_{jt}^c is a time-varying demand shock in destination j .

For the relative-share equation, the relevant instrument is the commodity-pair relative shifter

$$IV_{it}^{cc'} = \frac{IV_{it}^c}{IV_{it}^{c'}}. \quad (27)$$

Why it works. Counties that historically sell more of commodity c to destinations whose demand expands should face larger positive demand shocks to farm-gate prices for c . The identifying assumption is that historical exposure is uncorrelated with **changes** in unobservable relative supply shocks.

4.3 Identification of across-nest substitution (deforestation)

The agriculture-vs.-forest equation uses an aggregate agricultural demand shifter,

$$IV_{it}^C = \sum_{c \in C} IV_{it}^c. \quad (28)$$

Why it works. This instrument shifts the attractiveness of agriculture overall, without relying on direct observations of the unobservable payoff to natural land use A_{it}^N .

Key empirical conclusion. Frontier interactions in table 3 are negative, so both the commodity-switching elasticity and the deforestation elasticity are lower in frontier zones.

4.4 Identification of demand elasticities

The demand equations use **origin-side weather shocks** interacted with historical sourcing shares.

- For county-level sourcing, the instrument is a county-to-destination exposure term based on origin weather shocks.
- For nation-level substitution, the same idea is aggregated to the origin-nation level.
- For commodity-level substitution, the weather-shock exposure is aggregated across origin nations for each destination-commodity pair.

Why it works. Weather shocks move origin supply conditions, hence prices, but are plausibly orthogonal to time-varying demand residuals in specific destination markets after fixed effects.

4.5 Why the markdown geography is credible

The key markdown formula combines:

- the **estimated supply elasticity** of farmers,
- the **observed number of downstream buyers**, and
- a specific conduct assumption for agribusiness firms.

This means the paper’s geographic pattern of markdowns is credible to the extent that:

- the supply elasticities are well identified,

- local buyer counts are measured correctly,
- the oligopsony model is a reasonable approximation.

The paper also validates model-implied markdowns against external accounting markdowns and finds they move in the same direction across space.

Interpretation. The validation is not a clean causal test, but it is a useful external consistency check.

4.6 What is calibrated versus estimated

The model mixes estimated, observed, and calibrated objects:

- **Estimated:** supply substitution parameters $(\theta/\lambda, \lambda)$ and demand elasticities (η_l, η_m, η_u) .
- **Observed in data:** land endowments, population, nonagricultural wages, and several prices/quantities.
- **Taken from literature:** factor shares and the labor-supply elasticity across sectors.
- **Calibrated:** unobservable land productivity terms and trade-cost components used in equilibrium simulation.

Interpretation. The paper is a serious empirical model, but not every parameter is estimated inside one unified likelihood. Readers should treat the counterfactuals as disciplined quantitative exercises rather than purely nonparametric causal estimates.

4.7 Limitations to keep in mind

- The enforcement cost of targeted upstream policy is **not estimated directly**; it is represented in stylized form.
- The tracing problem for downstream certification is also modeled in a stylized way: the paper compares no targeting, regional targeting, and county targeting.
- Market power is modeled parsimoniously with symmetric intermediaries at each source location. Richer forms of conduct or bargaining are not the focus.
- The quantitative results depend on the estimated spatial correlation between emissions intensity and supply inelasticity; if that correlation were different in another industry, the policy ranking could change.

Interpretation. The paper's main message is qualitative but sharp: whether downstream environmental policy works depends crucially on how preexisting distortions line up with the externality.

5 Tables and figures

5.1 Figure 1: the externality is upstream and spatially heterogeneous

Read:

- The left panel shows that agricultural emissions come largely from **land use change** and other **upstream** sources rather than from downstream consumption.

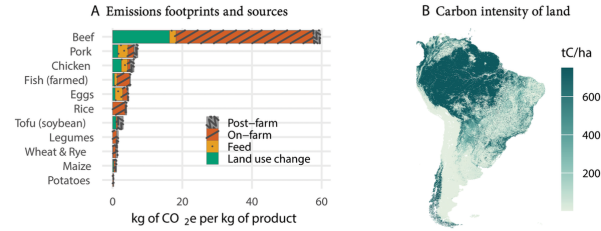


FIG. 1.—Sources of agricultural emissions along the supply chain. A, Emissions footprints (global averages) using publicly available data from Poore and Nemecek (2018). See appendix figure 10 (app. figs. 1–10 are available online) for footprints per kilocalorie or protein content. B, Aboveground carbon density constructed from biomass data from Spawn and Gibbs (2020).

- Beef has by far the largest emissions footprint among major products shown.
- The right panel shows that carbon stocks are extremely uneven across South America, with especially dense stocks in frontier forest regions.

Economic message. The policy problem is fundamentally different from fossil-fuel sectors. The emissions source is upstream and spatially heterogeneous, while feasible regulation is easier downstream.

5.2 Tables 1 and 2: a funnel-shaped supply chain and crude markdown evidence

TABLE 1
AGRIBUSINESS CONCENTRATION IN UPSTREAM MARKETS

	Brazil			Argentina	
	Beef	Maize	Soybean	Beef	Soybean
Number of agricultural establishments (sellers)	2,457,512	1,619,880	236,141	42,428	
Number of agribusiness firms (buyers)	134	110	188	32	
CR-1, national market	.34	.18	.15	.13	
CR-3, national market	.68	.46	.42	.36	
CR-1, median across local upstream markets	.66	1.00	1.00	.18	
CR-3, median across local upstream markets	.94	1.00	1.00	.43	
Share of upstream markets with only one agribusiness firm	.14	.86	.73	.01	

NOTE.—Upstream markets are defined at the county level and for 2017, the latest year. Agricultural establishment data are from censuses. All other data are from Trase. CR-N is the top-N firm concentration ratio.

TABLE 2
UPSTREAM MARKET CONCENTRATION AND ACCOUNTING-BASED MARKDOWNS

	Farm-Gate Price/Agribusiness Price Ratio (OLS)					
	(1)	(2)	(3)	(4)	(5)	(6)
CR-3	−.061* (.029)	−.062* (.031)	−.152*** (.022)	−.155*** (.023)	−.082*** (.022)	−.082*** (.021)
Market definition	Mesoregion	Mesoregion	Microregion	Microregion	AMC	AMC
Control for market access	×	✓	×	✓	×	✓
Observations	260	260	899	899	4,663	4,663

NOTE.—Regressions are at the upstream market level for various degrees of spatial granularity across columns. Farm-gate prices are from censuses and production surveys. Agribusiness prices are Trase FOB prices. CR-3 is the three-firm concentration ratio. A location's market access is its mean distance to surrounding hubs (weighted by hub size), defined as $MA_i = \sum_j s_j d_{ij}^{-1}$, where d_{ij} is distance between upstream market i and hub j and s_j is the hub's share of national exports. Heteroskedasticity-consistent robust standard errors are reported in parenthesis.

* $p < .10$.
*** $p < .01$.

Read:

- In Brazil there are about **2.46 million** cattle establishments but only **134** downstream beef-buying agribusiness firms.
- In the median local beef market, the **top firm** accounts for about **66%** of sourcing and the **top three firms** for about **94%**.
- For maize and soybean in many local markets, the top-three concentration ratio is essentially **1.00**.

Economic message. The paper's oligopsony channel is not a theoretical curiosity. The supply chain really is funneled through a very small number of buyers.

Read:

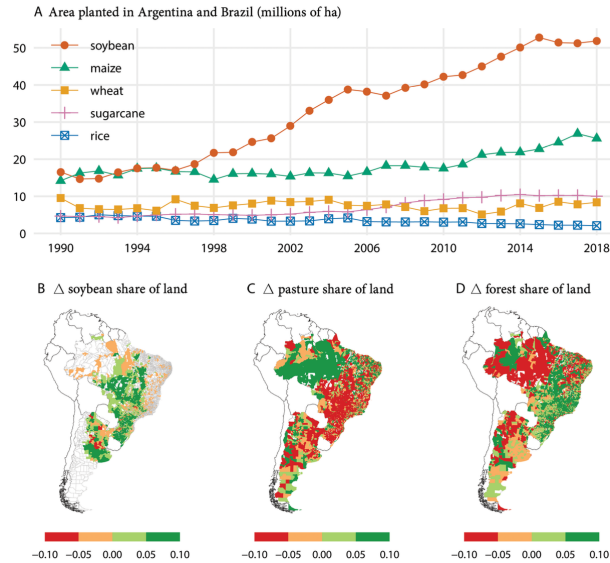


FIG. 2.—South American soybean boom across time and space. *A*, Total acreage allocated to major crops in Argentina and Brazil between 1990 and 2018. *B*, County-level changes in acreage allocated to soybeans, pasture, and forest between 1995 and 2017. Sources: Agricultural censuses, DA-MAGYP (Argentina), PAM (Brazil).

- Across all market definitions, the coefficient on **CR-3** is negative.
- In the finest market definition, the coefficient is about **-0.082** even controlling for market access.
- The simple farm-gate/FOB price ratio is therefore lower in places with more concentrated buyers.

Economic message. This table is not a structural causal estimate, but it is consistent with the paper’s market-power story: farm-gate prices are relatively compressed where buyer concentration is higher.

5.3 Figure 2: the soybean boom reallocates land across commodities and space

Read:

- Soybean acreage rises dramatically over time, from roughly the mid-teens of millions of hectares to above **50 million hectares** by the end of the sample.
- The maps show soybean land shares rising in core producing areas while pasture is displaced away from those areas.
- Forest shares decline especially in frontier regions where displaced cattle production expands.

Economic message. Environmental policy cannot be studied commodity by commodity in isolation. Comparative advantage and land competition make indirect land-use effects central.

5.4 Table 3 and Figure 3: frontier inelasticity generates wide markdowns where carbon is high

Read:

TABLE 3
SUPPLY-SIDE SUBSTITUTION PARAMETERS

	OLS (1)	IV (2)	OLS (3)	IV (4)
A. Within-Nest Substitution Elasticity: Estimate for Equation (7)				
θ/λ	.201*** (.034)	2.220*** (.144)	.164*** (.035)	2.219*** (.143)
$\theta/\lambda \times \text{frontier region}$			.725*** (.123)	-1.106*** (.180)
Time fixed effect	✓	✓	✓	✓
Location fixed effect	✓	✓	✓	✓
Observations	7,008	7,008	7,008	7,008
Kleibergen-Paap Wald first-stage F -statistic		171.2		176.6
B. Across-Nest Substitution Elasticity: Estimate for Equation (8)				
λ	.052*** (.009)	.279*** (.068)	.052*** (.010)	.281*** (.069)
$\lambda \times \text{frontier region}$			.002 (.031)	-.229** (.076)
Time fixed effect	✓	✓	✓	✓
Location fixed effect	✓	✓	✓	✓
Observations	9,255	9,255	9,255	9,255
Kleibergen-Paap Wald first-stage F -statistic		143.4		151.4

NOTE.—Standard errors are clustered at the county level.

** $p < .05$.

*** $p < .01$.

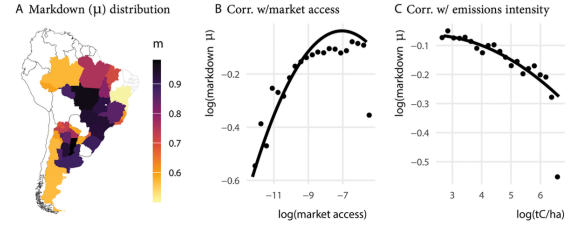


FIG. 3.—Spatial distribution of markdowns and correlation with local observables. All plots are constructed from county-level markdown estimates, market access terms, and carbon intensities. Markdowns are computed using equations (4) and (9) and averaged over time. Market access is constructed as $\sum_j s_j d_{ij}^{-1}$, where d_{ij} is distance between county i and hub j and s_j is hub trade volume. Carbon intensity is measured as tonnes carbon per hectare. For the scatterplots, curves are fitted polynomials of county-level observations, and markers are binned scatterplots. All results are for upstream beef markets. Additional results for soybean and maize are displayed in appendix C, section C.3.2.

TABLE 4
DEMAND SUBSTITUTION ELASTICITIES, DEPENDENT VARIABLE: LOG (expenditure share)

	Lower Level (across counties)		Middle Level (across nations)		Upper Level (across commodities)	
	OLS	IV	OLS	IV	OLS	IV
log (price)	1.266*** (.020)	-8.641*** (.691)	-1.478*** (.079)	-4.992*** (.772)	-1.030*** (.163)	-4.107** (1.173)
Kleibergen-Paap Wald first-stage F -statistic		257		32		9
Observations	409,558	409,558	38,272	38,272	4,059	4,059

NOTE.—Lower, middle, and upper specifications include origin nation–destination–year–commodity fixed effects, destination–year–commodity fixed effects, and destination–year fixed effects, respectively. Standard errors are clustered by origin–destination (for lower and middle levels) or destination (for upper level).

** $p < .05$.

*** $p < .01$.

- The IV estimate of the within-nest substitution parameter is about **2.220**.
- The frontier interaction is about **-1.106**, meaning frontier farmers switch across commodities much less easily.
- The IV estimate of the across-nest elasticity is about **0.281**, with a frontier interaction of about **-0.229**, so frontier deforestation is also less responsive to prices.

Economic message. The paper’s most important primitive is not concentration by itself, but **low supply elasticity in frontier regions**. That is what makes downstream power bite.

Read:

- Average model-implied markdowns are about **0.74** for beef and about **0.86–0.87** for maize and soybeans.
- Markdowns are wider in places with worse **market access**.
- Markdowns are also wider in places with higher **carbon intensity**.

Economic message. The two distortions line up in the worst possible way for downstream environmental policy: the Pigouvian signal is weakest exactly where emissions intensity is highest.

5.5 Table 4: final demand is elastic enough for trade diversion and reallocation

Read:

- The IV coefficient on log price is about **-8.641** at the county-within-nation level, implying a lower-level elasticity around **9.64**.
- The IV coefficient is about **-4.992** across source nations, implying $\eta_m \approx 5.99$.
- The IV coefficient is about **-4.107** across commodities, implying $\eta_u \approx 5.11$.

Economic message. Demand is not perfectly rigid. This matters because downstream taxes do not just depress one local market; they reallocate sourcing across counties, countries, and commodities.

5.6 Figures 4 and 5: the targeting-vs.-scale trade-off

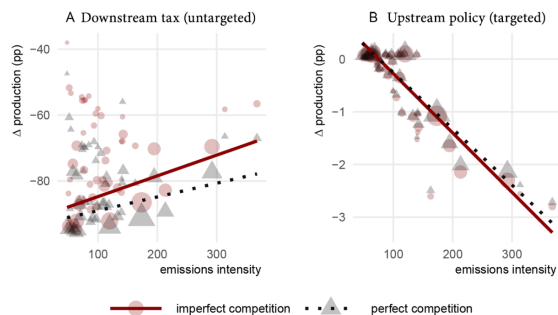


FIG. 4.—Targeting performance of downstream and upstream regulation. Scatterplots are constructed from upstream outcomes averaged to the state level and for the beef sector. Marker sizes are proportional to baseline production. Lines indicate the fit of a linear model to the markers, with observations size weighted by baseline production. The horizontal axis reports the emissions intensity of each upstream location (tonnes of CO₂e per tonne of output). The vertical axes report the percentage point change in upstream output as a result of each policy. Outcomes are reported for the default conduct assumption of imperfect competition and perfect competition.

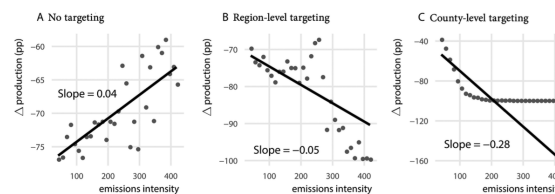


FIG. 5.—Coarse targeting flips the correlation between CO₂e intensity and abatement. All plots are constructed from county-level outcomes for the beef sector. Markers are binned scatterplots, and lines indicate fit to county-level outcomes. Horizontal axes report the emissions intensity of each upstream location (tonnes of CO₂e per tonne of output). Vertical axes report percentage point change in upstream output as a result of each policy. From left to right, results are shown for downstream taxes that are untargeted (A), coarsely targeted by upstream region (B), and finely targeted by upstream county (C). Outcomes are reported for the default conduct assumption of imperfect competition.

Read:

- In panel A, the untargeted downstream tax has the **wrong slope**: the output response becomes less negative as emissions intensity rises.
- In panel B, the targeted upstream policy has a strongly **negative slope**: it cuts production most in the dirtiest places.
- The gap between imperfect and perfect competition is much larger for the downstream tax than for the upstream policy.

Economic message. Downstream policy scales easily but transmits poorly; upstream policy targets correctly but is hard to scale. Market power mainly hurts the downstream option because it distorts pass-through.

Read:

- With **no targeting**, the slope between emissions intensity and abatement is positive.
- With **macroregion targeting**, the slope turns negative.
- With **county targeting**, the slope is even more negative.

Economic message. Certification does not need to be perfect to matter. Even coarse geographic certification goes a long way toward fixing the targeting problem of downstream taxes.

5.7 Figure 6 and Table 5: competition shapes policy transmission, and second-best policies recover only about one-third of the first best

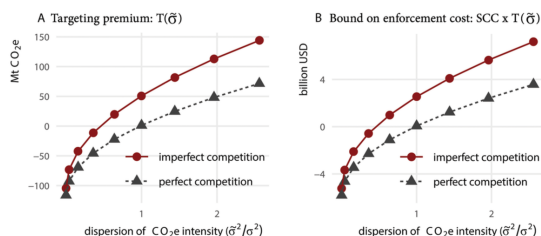


FIG. 6.—Interaction between emissions heterogeneity, market structure, and targeting. The vertical axis shows the targeting premium $T(\tilde{\sigma})$ for a given degree of emissions dispersion $\tilde{\sigma}$. *A* reports the targeting premium in units of CO_2e , while *B* reports its dollar value under a SCC of US\$50 per tonne. The horizontal axis in shows the deviation of the counterfactual dispersion $\tilde{\sigma}$ from the observed degree of dispersion σ , reported as the ratio $\tilde{\sigma}/\sigma$.

TABLE 5
DECOMPOSITION OF WELFARE IMPACTS

	Change Relative to Laissez-Faire Equilibrium (percentage points)					First-Best Effi- ciency Gains (%)	
	ΔE (1)	ΔPS (2)	ΔCS (3)	ΔCS_{SA} (4)	ΔCS_{ROW} (5)	ΔE (6)	ΔW (7)
A. Downstream Domestic Taxes							
Untargeted (sec. VI.A.1)	-26.88	-21.49	-28.07	-28.55	-27.57	29.27	26.40
Region targeted (sec. VI.A.3)	-30.48	-20.74	-26.36	-27.12	-25.60	33.20	30.85
County targeted (sec. VI.A.3)	-34.47	-18.96	-23.43	-24.26	-22.59	37.54	35.99
B. Downstream Foreign Taxes							
Unilateral EU tariff (app. D, sec. D.5.2)	-2.92	-2.44	-5.67	2.36	-13.80	3.18	2.44
C. Upstream Policies							
Forest subsidy (sec. VI.A.2)	-28.09	354.60	-1.10	-1.10	-1.10	30.60	32.55
First best (sec. VI.C.1)	-91.81	853.99	-37.62	-38.09	-37.14	100.00	100.00

NOTE.—Details on how each welfare component is computed are in app. D, sec. D.4. CS_{SA} refers to consumer surplus of South America, and CS_{ROW} refers to that of foreign consumers. Columns 1–5 display percentage point changes between the policy counterfactual and the laissez-faire. Columns 6 and 7 display the changes in emissions and overall welfare, as a percent of the changes delivered by the first best. Percentage point changes for government surplus are not displayed because the baseline level for G in the laissez-faire is zero (however, government surplus does indeed enter ΔW).

Read:

- The **targeting premium** rises as the dispersion of emissions intensity rises.
- The targeting premium is larger under **imperfect competition** than under perfect competition.
- The implied upper bound on society's willingness to pay for enforcement also rises with emissions heterogeneity.

Economic message. Market power does not mechanically help or hurt environmental policy. What matters is whether it raises or lowers pass-through in the dirtiest places. Here it lowers it, so competition improves transmission.

Read:

- The **untargeted downstream domestic tax** achieves about **29.27%** of first-best emissions reductions and **26.40%** of first-best welfare gains.
- **Region-targeted** and **county-targeted** downstream taxes improve to about **33.20% / 30.85%** and **37.54% / 35.99%**, respectively.
- The **targeted upstream forest subsidy** achieves about **30.60%** of first-best emissions gains and **32.55%** of first-best welfare gains.
- A **unilateral EU tariff** performs very poorly, with only about **3.18%** of first-best emissions gains and **2.44%** of welfare gains, reflecting leakage.

Economic message. There is no miracle second-best instrument. But better targeting—especially better downstream targeting—substantially improves both efficiency and distributional performance.

6 Short summary

- The paper studies environmental policy in a supply chain where emissions are generated upstream by farmers but policy is easiest to implement downstream on agribusiness firms.
- It builds a quantitative equilibrium model of South American agriculture with two key distortions: **environmental externalities** and **agribusiness monopsony power**.
- The core empirical result is that frontier regions are both **more emissions intense** and **less supply elastic**.
- Because markdowns are wider when supply is inelastic, market power is strongest in exactly those dirty frontier regions.
- A uniform downstream tax therefore transmits poorly: it reduces production least where environmental damage is highest and hits poor farmers hardest.
- A targeted upstream policy performs much better on targeting but is limited by enforcement cost and scale.
- Downstream policies can be improved a lot with **coarse certification** of origin.
- The sign of the correlation between market power and emissions intensity is the paper's central sufficient statistic for whether market power helps or hurts environmental policy transmission.
- In this setting the correlation is positive, so market power worsens policy targeting.
- The feasible second-best policies the paper studies recover only about one-third of the first-best gains.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that when environmental policy must be applied **away from the source of the externality**, the structure of the intervening supply chain becomes first-order. In South American agriculture, downstream agribusiness taxes do reduce emissions, but they do so inefficiently because the Pigouvian signal is transmitted through locations where supply elasticities and buyer power vary sharply. Since the dirtiest locations are also the least elastic and most exposed to buyer power, feasible downstream policies are badly targeted.

7.2 Economic intuition

The intuition is simple but powerful. A first-best environmental tax changes the incentives of the agents making the environmentally relevant decisions. But when regulation is imposed downstream, the incentive must travel back upstream through prices. If intermediaries have market power, they absorb part of the shock. If they absorb *more* of it in dirty places than in clean places, then the environmental signal is distorted in exactly the wrong direction. That is what happens here.

The paper also adds an important nuance to the classic idea that market power may reduce environmentally harmful output. In levels, oligopsony can indeed depress production and preserve some forest cover. But for the *transmission of corrective policy*, market power may help or hurt depending on where it is strongest. That depends on the empirical correlation between market power and the externality.

7.3 Takeaway

For environmental policy in fragmented developing-country supply chains, the main lesson is not simply “tax downstream” or “regulate upstream.” It is that **targeting, pass-through, and market structure must be designed jointly**. If first-best source-based regulation is infeasible, then coarse origin certification and other tools that preserve downstream scalability while improving upstream targeting may be the most promising middle ground.